Computer Architecture & CPU

A Deep Dive into the Foundations of Modern Computing Systems

Part 1: Architecture Overview

From Instruction Sets to System Integration

The Three Pillars of Architecture

Instruction Set (ISA)

The abstract interface between hardware and software. It defines instructions, registers, memory addressing modes, and native data types.

Microarchitecture

The physical implementation of the ISA. This includes pipeline design, cache hierarchy configuration, and execution unit organization.

System Architecture

The macroscopic integration of components. It encompasses memory subsystems, I/O interfaces, and data interconnects across the board.

Historical Architectural Milestones

Era	Major Architectural Breakthroughs
1960s	IBM System/360 introduced family compatibility; microprogramming emerged; CDC 6600 pioneered out-of-order execution concepts.
1980s	The RISC movement gained traction (Berkeley RISC, MIPS); x86 architecture gained dominance; pipelining became mainstream.
2000s	The "Power Wall" halted clock scaling; industry transitioned to multi-core processors to sustain performance growth.
2020s	Rise of heterogeneous computing, chiplet architectures (AMD Zen), domain-specific accelerators (NPUs), and RISC-V.

RISC vs. CISC Architectures

RISC (Reduced)

Focuses on fewer, simpler instructions executing in a single clock cycle.

- Fixed instruction length (typically 32 bits).
- Load-store architecture (only these access memory).
- Highly power-efficient (ARM, RISC-V).
- Lower code density requiring more instructions.

CISC (Complex)

Focuses on executing complex, multi-step operations via single instructions.

- Variable instruction length (1 to 15 bytes).
- Register-memory architecture (ALU operations on memory).
- Higher power consumption per instruction (x86).
- Higher code density compressing functionality.

Pipeline Processing & Execution

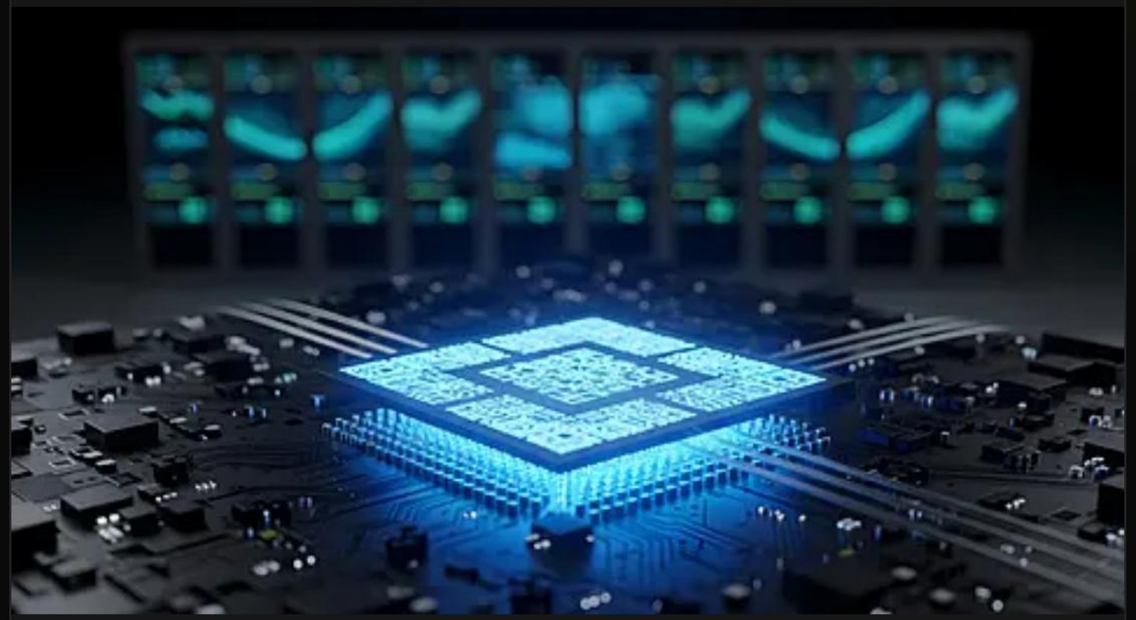
Overlapping Execution

A pipeline divides instruction execution into discrete stages (Fetch, Decode, Execute, Memory, Writeback), enabling simultaneous processing.

Superscalar Processors

Modern architectures issue multiple instructions per cycle by:

- Providing multiple redundant execution units (ALUs, FPUs).
- Implementing wide fetch and decode logic.
- Maintaining multiple issue slots for parallel dispatch.



Virtual Memory Parameters

4 KB

Base Page Size

Decoupling Addressing

Virtual memory isolates processes and simplifies management. The Page Table maps virtual to physical pages, handled in granular chunks (typically 4 KB, scaling up to 1 GB for huge pages).

A **Page Fault** occurs as a trap when accessing an unmapped page, invoking the OS handler to load the necessary data from secondary disk storage into main memory.

Part 2: The CPU Deep Dive

Execution Models, Parallelism, and Performance

The Instruction Lifecycle

1. Fetch Stage

The instruction address is pulled from the Program Counter (PC), retrieved from the instruction cache, and fetched into the pipeline.

2. Decode Stage

Instruction bits are translated to identify the opcode. Operands are read concurrently from the processor's register file.

3. Execute Stage

The Arithmetic Logic Unit (ALU) performs the designated mathematical operation, bitwise logic, or address calculation.

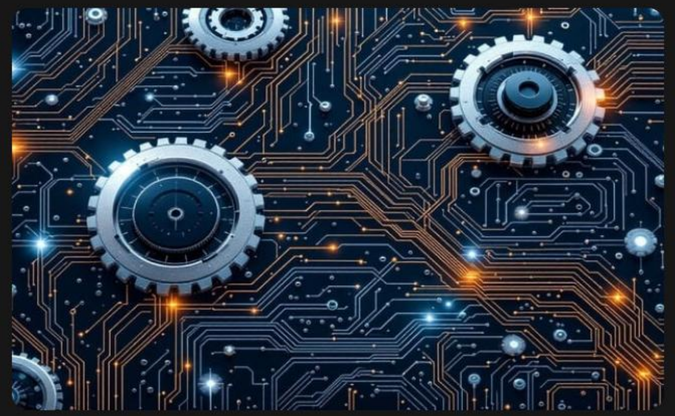
4. Memory & Writeback

Data caches are accessed for load/store commands. Finally, the computed result is written back to the register file or commit state.

Fundamental CPU Performance Factors

Instruction Count

Driven by ISA, compiler optimizations, and algorithm choice. Optimized via loop unrolling and function inlining.



Cycles Per Instruction

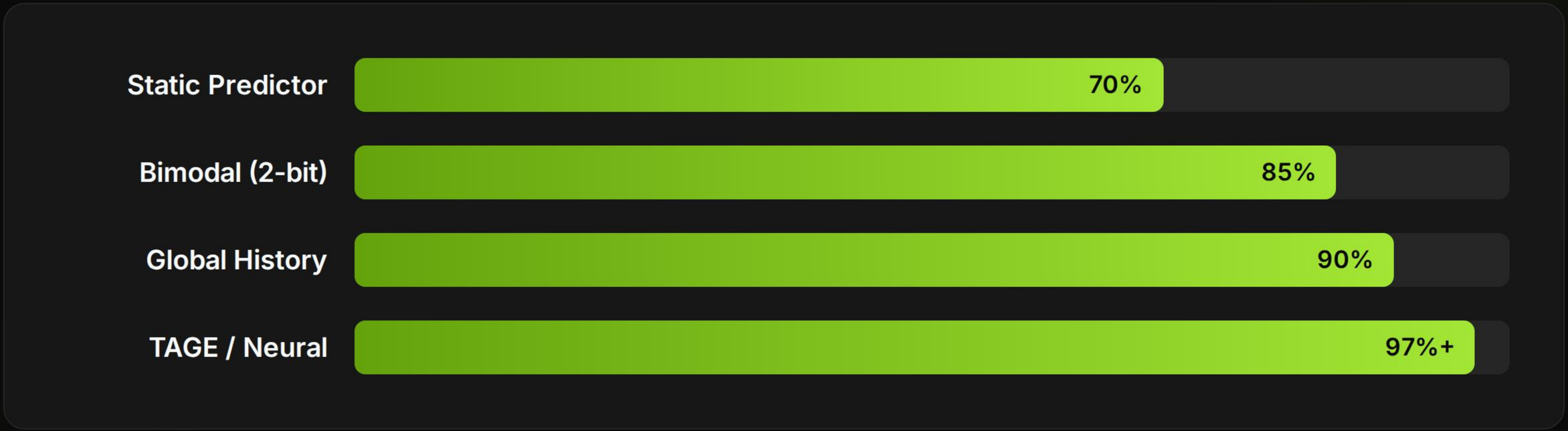
Influenced by pipeline stalls, cache misses, and branch mispredictions. Solved via out-of-order execution.

Clock Frequency

Determined by process technology and pipeline depth. Deeper pipelines enable higher operating frequencies.

$$\text{CPU Time} = \frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Cycles}}{\text{Instruction}} \times \frac{\text{Seconds}}{\text{Cycle}}$$

Branch Predictor Accuracy Comparison



Advanced prediction mechanisms like TAGE and Perceptron neural networks dramatically reduce control hazard penalties, minimizing pipeline flushes caused by mispredictions.

The Future is Heterogeneous

Quantum Computing & Domain-Specific Accelerators

Any questions on CPU microarchitecture or ISA mechanics?

Image Sources

https://png.pngtree.com/thumb_back/fh260/background/20251031/pngtree-futuristic-glowing-cpu-chip-illuminated-with-vibrant-blue-and-green-light-image_20139303.webp

Source: pngtree.com

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